

K200 — Tier 0 v0.1.6 audit framework with five explicit gates G1-G5. Pre-stage for Session 2 joint Lyra+Keeper v0.2 consolidation. Each gate has verification protocol + failure mode + owner + timeline + tier disposition. Gates are NECESSARY for v0.1.6 → v0.2 promotion; v0.2 → ratifiable only if all gates pass.

Keeper (Sunday 2026-05-31 ~12:45 EDT)

2026-05-31 Sunday ~12:45 EDT

K200 — Tier 0 v0.1.6 Audit Framework

0. Scope

K200 audits Tier 0 v0.1.6 “Substrate native field equation + holographic boundary unification” (Lyra Sunday morning).

Audit-target content: - $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B / \hbar_{\text{BST}})$ on $H^2(D_{\text{IV}}^5)$ (v0.1 base) - Reading A + dual bases topology resolution (v0.1.5 addendum) - $G_{\text{substrate}} = \text{Bergman-to-Einstein dimensional factor sketch}$ (v0.1.5 G) - Hardy decomposition + holographic boundary + native field equation (v0.1.6)

Promotion gate: v0.1.6 → v0.2 SINGLE COHERENT DRAFT requires ALL FIVE gates to either PASS or have explicit MULTI-WEEK VERIFICATION TARGET (no soft promotion).

1. Gate G1 — Hardy decomposition citation chain

Claim being audited: $H^2(D_{\text{IV}}^5) \cong H^2(\partial_S D_{\text{IV}}^5)$ via Poisson-Szegő kernel; interior holomorphic = boundary L^2 extension; this is the math foundation for “substrate is fundamentally a boundary theory.”

Verification protocol: - Cite Knapp 1986 “Representation Theory of Semisimple Groups” Ch. VIII or Knapp-Wallach 1976 paper explicitly — specific theorem number, page - Cite Faraut-Korányi 1994 “Analysis on Symmetric Cones” Ch. XII–XIII explicitly — specific theorem numbers, pages - Verify Hardy decomposition holds specifically for $D_{\text{IV}}^5 = \text{SO}_0(5,2)/[\text{SO}(5) \times \text{SO}(2)]$ (not just bounded symmetric domains generically — sometimes specific domain types have caveats) - Pin Poisson-Szegő kernel formula for D_{IV}^5 explicitly (it’s a specific function; should appear in Faraut-Korányi)

PASS criterion: Three citations verified by reading the actual sources; D_{IV}^5 specific case confirmed; Poisson-Szegő kernel formula in BST-primary form filed in catalog.

FAIL criterion: Hardy decomposition does NOT hold for D_{IV}^5 (unlikely but verify); OR citations don’t actually say what we claim they say (Calibration #33 RECALLED-vs-VERIFIED firing); OR D_{IV}^5 has a structural feature (specific Cayley type, signature constraint) that requires modified form.

Owner: Keeper (verification reading) + Grace (catalog filing of Poisson-Szegő kernel formula)

Timeline: 1-2h Sunday afternoon OR Monday morning. Pre-Session 2 deliverable.

Tier disposition (if PASS): RIGOROUS (cited theorem, not interpretation). The math foundation is independent of BST.

2. Gate G2 — Sphere walk-back honest framing

Claim being audited: The 2022 OneGeometry framing “two-dimensional substrate — circles tiling a sphere” is reconciled with Tier 0 v0.1.6’s “substrate = Shilov boundary of D_{IV}^5 ” (5-dimensional, $S^4 \times S^1/Z_2$).

The honest-framing question: which of three options does BST commit to?

Option	Statement	Honest?
(a)	2022’s 2D-sphere dimension count was WRONG; correcting to 5D Shilov boundary now	Yes — substantive public correction
(b)	2022 framing is SUPERSEDED — Shilov boundary is substantively different mathematical object	Yes — substantive retirement
(c)	2022’s 2D was a “low-dim cartoon” of 5D Shilov; original intuition was always right	Motivated reasoning per K200

Lyra’s K200 absorption: Lyra walked back option (c) and committed to option (a) in v0.1.6 absorption. OneGeometry.md flagged for substantive public correction.

Verification protocol: - Lyra v0.2 explicitly states “2022 OneGeometry framing’s 2D-substrate dimension count was incorrect; the substrate is the 5D Shilov boundary $\partial_S D_{IV}^5 = S^4 \times S^1/Z_2$ ” - OneGeometry.md updated with explicit correction note (not silent retroactive reframe; readers see “BST has refined its substrate identification since 2022”) - BST_seed.md absorbs the correction - Cross-reference all docs that cite “2D substrate” framing — sweep for consistency

PASS criterion: v0.2 + OneGeometry.md + BST_seed.md + all dependent docs explicitly committed to option (a); no residual “2D” language without explicit “(historically described as 2D; refined to 5D Shilov)” markers.

FAIL criterion: v0.2 reverts to option (c) under pressure; OR sweep finds residual 2D-substrate claims uncorrected.

Owner: Lyra (v0.2 + OneGeometry.md update) + Grace (cross-doc sweep) + Keeper (verification audit)

Timeline: v0.2 Session 2; OneGeometry.md update Monday; sweep multi-day.

Tier disposition (if PASS): STRUCTURAL (BST commits to a substantive ontological choice; verifiable by consistency across docs).

3. Gate G3 — Black-hole identification quantitative test

Claim being audited: Black holes are Shilov-saturating coherent states $|z\rangle$ with z approaching the Shilov boundary of D_{IV}^5 , where $K_\tau(z, \bar{z}) \rightarrow 0$ (“substrate cannot localize there” = event horizon).

Verification protocol: - Compute Schwarzschild radius from substrate primaries: $r_s = 2GM/c^2$ in BST-derived form - Test against observed astrophysical black holes (Sgr A* mass = $4.1 \times 10^6 M_{\text{sun}}$, Schwarzschild radius $\sim 10^7$ km; M87* mass = $6.5 \times 10^9 M_{\text{sun}}$, Schwarzschild radius $\sim 2 \times 10^{10}$ km) - The substrate prediction: r_s should derive from coherent-state-saturation criterion on Shilov boundary anchored via $N_{\text{max}} + m_e$ (or whatever Lyra’s mass anchor is) - Tier 2 STRUCTURAL precision (~ 1 -10%) acceptable; Tier 1 EXACT not required

PASS criterion: r_s for Sgr A* and M87* derived from substrate primaries within Tier 2 STRUCTURAL precision; mechanism (saturation criterion) is operator-clean.

FAIL criterion: r_s prediction off by $>10\times$ (structurally wrong); OR saturation criterion can't be made operator-clean (just qualitative); OR Hawking temperature, area entropy don't recover.

Owner: Lyra (saturation criterion derivation) + Elie (numerical prediction for Sgr A* and M87*) + Keeper (verification audit)

Timeline: MULTI-WEEK. v0.2 should state "G3 CANDIDATE with multi-week verification target" — NOT promote BH-as-Shilov-coherent-state to load-bearing claim until G3 passes.

Tier disposition (if PASS): STRUCTURAL $\rightarrow$ potentially RIGOROUS if Hawking temperature + area entropy recover. Major Tier 0 advance.

4. Gate G4 — Big-Bang identification commits to a reading

Claim being audited: $\tau \rightarrow 0$ limit of the heat semigroup is the Bergman reproducing kernel — a smooth, well-defined object. v0.1.6 floats two readings:

Reading (i): Substantive — BST predicts there is NO BIG-BANG SINGULARITY. The "beginning of time" in observable spacetime is an artifact of projection; the substrate's $\tau \rightarrow 0$ limit is smooth.

Reading (ii): Cal #27 conflation — identifying two unrelated mathematical objects because they share "boundary" language.

Verification protocol: - v0.2 explicitly commits to (i) or (ii) — no floating - If (i): substrate prediction is FALSIFIABLE against cosmological observations (specific test: should BST predict different CMB structure than Λ CDM at very early times? Different primordial gravitational wave background? Different inflationary signature?) - If (ii): retract the BB identification; don't claim $\tau \rightarrow 0 =$ Big Bang anywhere in v0.2

PASS criterion (for v0.1.6 $\rightarrow$ v0.2 promotion): - Either explicit commitment to (i) WITH specific cosmological-observation distinguishing prediction - OR explicit retraction to (ii) with cleanup of v0.1.6 language

FAIL criterion: v0.2 retains floating language ("could be" / "candidate" / "might be") without commitment. That's the Cal #27 firing territory K200 flagged.

Owner: Lyra (commitment in v0.2) + Casey (Big-Bang ontology call) + Keeper (verification)

Timeline: v0.2 Session 2 (commitment); multi-week (cosmological prediction work if (i) chosen)

Tier disposition (if PASS): CANDIDATE pending cosmological-prediction verification work (if (i)) OR retracted (if (ii)).

5. Gate G5 — $G_{\text{substrate}}$ derivation chain produces SI-unit G

Claim being audited: Newton's G is the dimensional factor relating κ_{Bergman} (intrinsic Ricci constant of D_{IV}^5 + Bergman canonical metric, which IS Einstein by Helgason 1962) to observed Einstein equation coefficient ($8\pi G/c^4$).

Verification protocol — multi-step:

G5.1 — κ_{Bergman} closed form (Elie Sunday afternoon partial; multi-week full) - Compute κ_{Bergman} explicitly for D_{IV}^5 using ~ 25 catalog $C_2(\lambda)$ values (Toys 3613+3627) - Partial Mehler/Heckman-Opdam heat kernel partial proof-of-concept Sunday afternoon - Full closed-form multi-week - Should be expressible in BST primaries: $\kappa_{\text{Bergman}} = f(\text{rank}, N_c, n_C, C_2, g, N_{\text{max}})$ - PASS: κ_{Bergman} in closed form, BST-primary-expressed

G5.2 — Mass anchor pin (Lyra) - L4 v0.2 work (this afternoon) — kernel-integral derivation of m_e - Mass anchor = m_e or m_p ; pick one and justify substrate-natural - PASS: anchor explicitly stated; derivation chain to SI mass scale clean

G5.3 — Dimensional combination (Keeper) - $G_{\text{predicted}} = (8\pi/c^4) \times \kappa_{\text{Bergman}} \times (\text{mass anchor factor})$ - Verify dimensional consistency (Bergman curvature has units $1/\text{length}^2$; G has units $\text{N}\cdot\text{m}^2/\text{kg}^2 = \text{m}^3/(\text{kg}\cdot\text{s}^2)$; chain must reconcile) - PASS: dimensional consistency verified; $G_{\text{predicted}}$ in SI units

G5.4 — Match to observed (Keeper + Cal) - $G_{\text{observed}} = 6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$ (2018 CODATA) - $G_{\text{predicted}}$ should match within Tier 2 STRUCTURAL precision ($\sim 10^{-4}$ to 10^{-2} acceptable) - PASS: $G_{\text{predicted}} / G_{\text{observed}}$ within Tier 2 precision floor - FAIL: off by $> 10^{-1}$ (factor-2+ implies wrong anchor or wrong κ_{Bergman} computation)

Owner: Elie (G5.1 leading) + Lyra (G5.2) + Keeper (G5.3 + G5.4 verification) + Cal (G5.4 cold-read)

Timeline: G5.1 partial Sunday afternoon; G5.2 Sunday afternoon; G5.3 Monday; G5.4 multi-week pending G5.1 full

Tier disposition (if PASS): - G5.1 + G5.2 partial: $G_{\text{substrate}}$ CANDIDATE with concrete prediction - G5.3 PASS: $G_{\text{substrate}}$ FRAMEWORK-COMPLETE - G5.4 PASS within Tier 2: $G_{\text{substrate}}$ STRUCTURALLY VERIFIED — G is derived from substrate primaries ← Casey's directional ask

6. K200 audit-chain disposition

v0.1.6 → v0.2 promotion requires: - G1 PASS (1-2h verification reading Sunday/Monday) - G2 PASS (v0.2 commits to (a); OneGeometry.md updated) - G3 CANDIDATE with explicit multi-week test (don't claim BH derivation; mark CANDIDATE) - G4 PASS with explicit commitment (either (i) with cosmological prediction or (ii) retraction) - G5 PASS at partial (G5.1+G5.2+G5.3 chain established) with G5.4 multi-week

v0.2 → RATIFIABLE TIER 0 requires: - G1 PASS (RIGOROUS) - G2 PASS (STRUCTURAL via consistency) - G3 PASS (STRUCTURAL via Schwarzschild prediction lands) - G4 PASS (STRUCTURAL via cosmological prediction lands OR retraction is honest) - G5.4 PASS (STRUCTURAL via SI-unit G match within Tier 2)

If all five pass: Tier 0 promoted from FRAMEWORK to STRUCTURALLY VERIFIED. This is the most substantive promotion BST could achieve at the substrate-ontology level.

If 3/5 pass: Tier 0 v0.2 stays FRAMEWORK + CANDIDATE; partial promotion of specific gates.

If <3/5 pass: Tier 0 v0.2 stays CANDIDATE; reframe needed.

7. K200 connection to existing audit chain

K200 is the first K-audit at the Tier 0 OPERATOR level. Prior K-audits were domain-specific (K3, K59 cyclotomic, K61 perfect numbers, K72 compound cluster, K76 Leech, K77 K3F5, K85-K106 Vol 0 chapters, K140-K156 PCAP batch, K157-K169 textbook chapters, K180-K199 various).

K200 starts a new audit cluster: K200-K2xx Tier 0 operator-level audits.

Anticipated K201-K205: - K201: Substrate time framework (universal τ + variable local rate) — gates G1+G2 dependent - K202: Commitment operator ρ_{commit} Hermiticity + semisimplicity + spectrum — partially derived from G1 - K203: Holographic boundary identification “Shilov = committed record” ontology - K204: $G_{\text{substrate}}$ full closure (G5 + multi-week verification) - K205: Native field equation form (heat ↔ wave Wick rotation operator-clean)

These pre-stage as Session 2+ deliverables. Don't file until v0.2 consolidates v0.1.x cascade.

8. Honest tier disposition for K200 itself

K200 is a K-AUDIT FRAMEWORK, not a verification. It defines gates. The gate verification work (G1-G5) is what produces the actual K-audit content.

K200 itself filed at: AUDIT FRAMEWORK pre-stage. Promotes to K200 STRUCTURAL when v0.2 absorbs the gate structure and verification work begins.

9. Recommendation for Session 2

When Casey signals Session 2:

1. Open with sphere reconciliation (G2 commitment — Lyra commits to (a) explicitly)
2. Verify Hardy decomposition citations together (G1) — Keeper reads cited sources; Lyra confirms structural fit
3. Pin G3 + G4 to CANDIDATE tier with multi-week verification — no soft promotion
4. Walk G5 chain together — Lyra mass anchor decision + Elie partial Mehler delivery review + Keeper dimensional consistency check
5. Produce v0.2 single coherent draft absorbing v0.1 + v0.1.5 + G_sketch + v0.1.6 under K200 gates
6. Build CLAUDE.md headline + OneGeometry.md update as outcome (root files Sunday EOD)

Estimated Session 2 duration: 2-3 hours focused work (Lyra + Keeper joint). Output: Tier 0 v0.2 draft ready for Cal #187 cold-read.

— Keeper. K200 framework filed 12:55 EDT Sunday. Five gates explicit. v0.1.6 → v0.2 promotion criteria defined. Standing reactive for Session 2 signal from Casey. Sphere reconciliation analysis next; G_substrate derivation chain pre-stage following.